

Array Problem-Solving Techniques

A reference guide for DSA interview preparation — Bhuvneshwar

Core Techniques (Must Know)

1. Two Pointers

Used on sorted arrays to find pairs/triplets. Example: Two Sum II, Container With Most Water.

2. Sliding Window

For subarray/substring problems. Example: Max Sum Subarray of Size K, Longest Substring Without Repeating Characters.

3. Prefix Sum

For range sum queries and subarray sum problems. Example: Subarray Sum Equals K.

4. Kadane's Algorithm

Special case of DP for maximum subarray sum. Example: Maximum Subarray.

5. Binary Search on Array

For sorted array search problems. Example: Search in Rotated Sorted Array.

6. Sorting-based Approach

Used when order doesn't matter. Example: Merge Intervals, 3Sum.

7. Hashing / HashMap

For frequency counting and duplicate detection. Example: Two Sum, Contains Duplicate.

8. Dutch National Flag Algorithm

3-way partitioning technique. Example: Sort Colors (0s, 1s, 2s).

9. Cyclic Sort

Used when array contains numbers in range 1 to N. Example: Find Missing Number, Find Duplicate Number.

10. Fast & Slow Pointers (Floyd's)

Cycle detection technique, also applies to array-index-as-pointer problems. Example: Find Duplicate Number.

Intermediate / Advanced Techniques

11. Merge Intervals Technique

For overlapping interval problems. Example: Merge Intervals, Insert Interval.

12. Matrix Traversal Techniques

Spiral order, rotation, transpose. Example: Spiral Matrix, Rotate Image.

13. Divide and Conquer

Merge sort based approach. Example: Count Inversions.

14. Greedy Approach

Local optimal choice leading to global optimum. Example: Jump Game, Gas Station.

15. Dynamic Programming on Arrays

For subset sum, LIS, partition-type problems. Example: Longest Increasing Subsequence.

16. Monotonic Stack

For next greater/smaller element problems. Example: Next Greater Element, Stock Span.

17. Bit Manipulation

XOR-based tricks. Example: Single Number.

18. Binary Search on Answer

Search over the answer space rather than the array. Example: Allocate Books, Painter's Partition.

Suggested Learning Order: Master techniques 1–9 first (most frequently asked in interviews), then move to Trees and Graphs. After that, techniques 11–18 will be much easier to pick up as part of the DevOps + DSA placement roadmap.